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Deepfake Voice Detection

Graduation Project **1** prepared to obtain the degree of B.Sc. in the Department of
Artificial Intelligence Engineering and Data Science

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Abstract

Voice deepfake detection has become one of the most critical challenges in the fields of cybersecurity and speech processing due to the rapid advancement of artificial intelligence techniques capable of generating highly realistic synthetic speech. The danger of this technology lies in its potential misuse in identity theft, fraud, misinformation, and bypassing voice authentication systems. Furthermore, detecting fake speech is a challenging task because of the high similarity between genuine and synthesized voices, the diversity of speech generation methods, and variations in audio quality and recording conditions.

This project aims to develop an intelligent voice deepfake detection system using deep learning techniques. The proposed approach relies on preprocessing audio signals and converting them into Mel-Spectrogram representations, followed by a hybrid CNN-LSTM architecture. Convolutional Neural Networks (CNNs) are utilized to extract spectral features and artifacts from audio signals, while Long Short-Term Memory (LSTM) networks are employed to capture temporal dependencies and sequential speech patterns. The preprocessing stage includes converting audio to mono channels, resampling signals to 16 kHz, silence removal, and normalization to ensure consistent input quality. The main strength of the proposed solution lies in combining both spectral and temporal analysis, which enhances the model's ability to identify subtle abnormalities generated by deepfake synthesis techniques. In addition, advanced evaluation metrics such as Recall, F1-Score, and Equal Error Rate (EER) are adopted to provide a comprehensive assessment of the model performance.

The experimental results demonstrated that the proposed model achieved promising performance in distinguishing between genuine and fake speech samples. The training process showed stable convergence with decreasing loss values and improvements in evaluation metrics. Moreover, preprocessing techniques and fixed-length Mel-Spectrogram representations contributed to improving model stability and reducing overfitting. The findings indicate that combining Mel-Spectrogram features with a CNN-LSTM architecture is an effective approach for voice deepfake detection tasks. Future work may involve integrating self-supervised learning models such as Wav2Vec2 to further improve generalization capability and robustness against unseen spoofing techniques.

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Chapter One: Introduction :

1.1 Introduction :

Artificial Intelligence (AI) has become one of the most influential technologies in modern life, contributing to major advancements in areas such as healthcare, education, cybersecurity, communication, and multimedia processing. Among the most important applications of AI is speech processing, which enables machines to analyze, understand, and generate human speech with remarkable accuracy. Recent developments in deep learning have significantly improved the quality of synthesized speech, making artificial voices increasingly realistic and difficult to distinguish from genuine human speech.

One of the emerging technologies in this field is known as Deepfake technology, which uses advanced neural networks to generate synthetic media, including fake images, videos, and voices. In the audio domain, voice deepfakes are capable of imitating a person's voice characteristics, speaking style, and tone with high precision. Although these technologies can provide useful applications such as virtual assistants, speech restoration, and entertainment systems, they also introduce serious security and ethical concerns.

The misuse of voice deepfake technology may lead to identity theft, financial fraud, misinformation, and unauthorized access to voice-based authentication systems. As synthetic speech generation methods continue to improve, distinguishing between real and fake speech becomes increasingly challenging. Therefore, developing reliable systems capable of detecting manipulated or synthesized speech has become an important research topic in artificial intelligence and cybersecurity.

This project focuses on the field of voice deepfake detection by exploring deep learning techniques for analyzing audio signals and identifying differences between genuine and fake speech recordings. The proposed work aims to contribute to improving digital security and enhancing trust in voice-based communication systems through intelligent audio analysis methods.

1.2 Research Problem :

The scientific problem addressed in this research is the increasing difficulty of distinguishing between genuine speech and fake speech generated by modern deepfake technologies. Recent AI-based speech synthesis models are capable of producing highly realistic voices that closely mimic human speech characteristics, making traditional detection methods ineffective in many cases. This problem becomes more challenging due to the diversity of speech generation techniques, variations in recording quality, background noise, and acoustic environments, in addition to the strong similarity between genuine and synthesized

speech patterns. Furthermore, voice deepfake detection systems face significant challenges in extracting discriminative features that can capture subtle differences between real and fake audio signals while maintaining strong generalization capability against unseen attacks and datasets. Therefore, the core research problem lies in designing an intelligent model capable of analyzing audio signals and extracting meaningful spectral and temporal features to accurately and reliably detect voice deepfakes

1.3 Research Objectives :

This research aims to develop an intelligent system for voice deepfake detection using deep learning and audio signal processing techniques to accurately and reliably distinguish between genuine and fake speech recordings. The proposed system relies on preprocessing audio signals and converting them into Mel-Spectrogram representations, followed by a hybrid CNN-LSTM architecture in which CNN networks are used to extract spectral features while LSTM networks are employed to capture temporal dependencies within speech signals. The research also aims to improve the model's ability to detect abnormal patterns and artifacts generated by synthetic speech techniques while achieving a classification accuracy of at least 90% and reducing the Equal Error Rate (EER) to obtain more reliable performance. In addition, the study seeks to evaluate the proposed model using multiple metrics such as Accuracy, Recall, F1-Score, and EER, as well as reduce overfitting and improve generalization capability through preprocessing and data handling techniques. Furthermore, the research investigates the future integration of self-supervised learning approaches to enhance the robustness and effectiveness of the system against different types of voice spoofing attacks

1.4 Research Contribution :

The contribution of this research lies in developing an intelligent voice deepfake detection system through the integration of audio signal processing techniques with deep learning models using a hybrid CNN-LSTM architecture. The proposed approach benefits from the capability of CNN networks to extract spectral features and audio artifacts, while LSTM networks are utilized to capture temporal dependencies and sequential speech patterns within audio signals. In addition, the project introduces a complete audio preprocessing pipeline that includes resampling, mono conversion, silence removal, and Mel-Spectrogram feature extraction, which improves the quality of extracted features and enhances training stability. Furthermore, the research emphasizes the use of advanced evaluation metrics such as Equal Error Rate (EER) alongside Accuracy and F1-Score to provide a more reliable assessment of system performance in voice spoofing detection tasks. The proposed framework also provides a scalable foundation for future development through the integration of self-supervised learning techniques such as Wav2Vec2 to improve generalization capability and robustness against unseen deepfake attacks

1.5 Beneficiaries of the Research :

Several sectors can benefit from this research due to the growing importance of voice deepfake detection in enhancing digital security and the reliability of modern communication systems. One of the main beneficiaries is security and governmental organizations that rely on voice-based verification services, where the proposed system can help detect identity impersonation and voice fraud attempts. Banks and financial institutions can also benefit from fake speech detection technologies to secure voice authentication systems and reduce financial fraud risks. In addition, telecommunication companies and social media platforms may utilize such systems to limit the spread of fake audio content and fraudulent calls. Research centers and universities are also considered important beneficiaries, as the project provides a practical framework that can be further developed in the fields of artificial intelligence, speech processing, and cybersecurity. Furthermore, the proposed system can be integrated into future voice assistant applications and biometric authentication systems to improve security and trust in digital services

Chapter Two: Theoretical Study :

This project on voice deepfake detection relies on several mathematical and computational theories related to audio signal processing, deep learning, and temporal data analysis. These theoretical foundations are essential for understanding how audio signals are processed and transformed into representations suitable for neural network analysis, as well as how the proposed model learns patterns that distinguish genuine speech from synthesized speech.

The theoretical study begins with the concept of digital audio signals, where speech is represented as a sequence of digital samples extracted from the original analog signal at a specific sampling frequency known as the Sample Rate.

This rate determines the number of samples captured per second and directly affects the quality and precision of audio representation. In this project, a sampling rate of 16 kHz was adopted because it provides a good balance between speech quality and computational efficiency for speech processing tasks.

The project also relies on audio signal processing techniques to transform raw audio into representations more suitable for deep learning applications. One of the most important techniques is converting audio signals into spectral representations such as Spectrograms and Mel-Spectrograms. This process is based on the Fast Fourier Transform (FFT), which converts audio signals from the time domain into the frequency domain to analyze the frequencies contained within speech. Since speech characteristics change over time, the Short-Time Fourier Transform (STFT) is applied by dividing audio signals into small overlapping windows and analyzing each segment independently to extract both temporal and spectral information.

Furthermore, the project utilizes the Mel Scale, which is designed to simulate human auditory perception, where the human ear is more sensitive to lower frequencies than higher frequencies. Therefore, the frequencies obtained from STFT are projected into Mel frequency bands to generate Mel-Spectrogram representations. This representation transforms audio into a two-dimensional image describing the relationship between time, frequency, and signal energy. Mel-Spectrograms are widely used in speech processing and deepfake detection tasks because they allow neural networks to learn hidden patterns and synthesis artifacts within speech signals.

The proposed system also relies on deep learning techniques, particularly Convolutional Neural Networks (CNNs) and Long Short-Term Memory (LSTM) networks. CNNs are employed to extract local spectral features and frequency patterns from Mel-Spectrogram images due to their strong capability in analyzing two-dimensional data and detecting fine-grained patterns. On the other hand, LSTM networks are used to analyze long-term temporal dependencies within audio sequences, as they can preserve important information over time and address the Vanishing Gradient Problem commonly found in traditional recurrent neural networks.

In addition, the project is based on supervised learning principles, where the model is trained using labeled datasets containing both genuine and fake speech samples. The model learns discriminative patterns for each class through iterative optimization. Network

parameters are updated using the Backpropagation algorithm and the Adam optimizer to minimize the loss function and improve classification performance

Finally, the theoretical framework includes several statistical evaluation metrics such as Accuracy, Recall, F1-Score, and Equal Error Rate (EER). These metrics are essential for evaluating the reliability and effectiveness of the proposed model, especially in imbalanced datasets where genuine and fake samples are not equally distributed

Chapter Three Reference study :

Study [1] presented a comprehensive and systematic review of voice deepfake detection approaches based on spectral features and deep learning techniques, with particular emphasis on Convolutional Neural Networks (CNNs) and modern architectures used in speaker verification systems. The study aimed to enhance the reliability and security of voice authentication systems used in sensitive applications such as banking services, intelligent assistants, and remote identity verification systems

The research focused on the growing threat caused by recent advances in Text-To-Speech (TTS) and Voice Conversion (VC) technologies, which are now capable of generating highly realistic synthetic speech that closely mimics human voices. These developments pose significant risks to voice-based security and authentication systems

The study adopted a systematic review methodology covering research published between 2020 and 2024. A large number of papers from academic databases such as IEEE, Springer, ACM, and ScienceDirect were analyzed and categorized according to the type of spectral features used, deep learning architectures, spoofing attack types, and evaluation metrics such as Equal Error Rate (EER) and t-DCF

Several spectral feature extraction methods were discussed, including Mel-Spectrogram, MFCC, and CQCC. The results showed that Mel-Spectrogram representations are highly effective for capturing temporal and spectral speech characteristics, making them well-suited for CNN-based architectures. In addition, CQCC features achieved superior performance when combined with ResNet architectures, particularly for detecting low-frequency artifacts associated with spoofed speech

The study reviewed both traditional and deep learning-based models, including GMM, PLDA, CNN, ResNet, and CNN+GRU architectures. Experimental findings demonstrated that discriminative deep learning models significantly outperformed traditional generative approaches in voice spoofing detection tasks

One of the best reported results in the study was achieved using a ResNet model combined with CQCC features on the ASVspoof2019 dataset, reaching an Equal Error Rate (EER) of 0.59%, which highlights the effectiveness of combining advanced spectral representations with modern neural architectures for deepfake voice detection

The main strength of this study lies in its comprehensive comparative analysis of recent research and its detailed discussion of current challenges in the field, including generalization limitations, noisy real-world environments, and restricted training data diversity. However, the study mainly focused on theoretical analysis and literature review without proposing a novel practical model or a complete implementation framework. Study [2] investigated the rapid advancements in deep learning-based speech generation technologies, particularly Text-To-Speech (TTS), Voice Conversion (VC), and modern generative architectures such as GANs and Diffusion Models. These technologies have become capable of generating highly realistic synthetic speech that is difficult to distinguish from genuine human voices. The study focused on the impact of these developments on speaker verification systems, voice banking services, digital identity systems, and intelligent assistants

The research problem centered on the limited capability of traditional speaker verification systems to differentiate between genuine speech and AI-generated fake speech, especially in noisy real-world environments or when encountering spoofing attacks that were not included in the training data. The study also highlighted the generalization problem caused by variations in language, speaking style, and the emergence of previously unseen speech generation models. The main objectives of the study were to analyze the most effective spectral features for voice deepfake detection, compare the performance of various deep learning architectures, evaluate model generalization capability, and investigate the impact of different spoofing scenarios such as Logical Access and Physical Access attacks. The study also proposed future research directions for improving spoofing detection systems.

The proposed methodology followed an experimental and analytical approach. Raw audio signals were transformed into several spectral representations including Mel-Spectrogram, MFCC, CQCC, Spectral Contrast, and Chroma Features in order to extract subtle characteristics that may distinguish synthetic speech from genuine speech. Several deep learning models were then evaluated, including CNN, ResNet, and hybrid architectures such as CNN+GRU and CNN+LSTM, in addition to traditional models such as GMM and SVM.

The study utilized benchmark datasets commonly used in spoofing detection research, including ASVspoof2019 and ASVspoof2021. These datasets contain both Logical Access attacks based on TTS and Voice Conversion techniques, and Physical Access attacks based on replay attacks.

Performance evaluation was conducted using standard metrics such as Equal Error Rate (EER) and Tandem Detection Cost Function (t-DCF), which are widely adopted in voice spoofing detection research. Experimental results showed that advanced spectral features such as CQCC outperformed traditional MFCC features. In addition, ResNet architectures achieved better performance than conventional CNN models in most scenarios.

The study also demonstrated that hybrid architectures such as CNN+GRU and CNN+LSTM improved temporal signal analysis and enhanced spoofing detection performance. However, the results revealed that model performance significantly degrades when tested on unseen spoofing attacks that were not included in the training phase, highlighting the persistent generalization challenge in modern voice deepfake detection systems.

One of the major strengths of this study is its comprehensive comparison of modern spectral features and deep learning architectures using standardized benchmark datasets, which increases the reliability of the reported results. However, the study mainly focused on experimental evaluation and did not propose a complete real-time deployable system capable of operating effectively in complex real-world environments. Study [3] investigated the problem of voice deepfake detection using Machine Learning techniques in response to the rapid spread of AI-based speech generation technologies capable of imitating human voices with high realism and accuracy. The study highlighted the growing threats posed by such technologies in sensitive areas including financial fraud, identity impersonation, misinformation dissemination, and attacks on speaker verification systems. Before the training phase, multiple preprocessing steps were applied, including noise removal, audio length normalization, and transforming speech signals into numerical representations suitable for machine learning algorithms. The models were then trained using labeled datasets containing both real and deepfake voice samples.

The study utilized the ASVspoof dataset, which is considered one of the most widely used benchmark datasets in voice spoofing detection research. The dataset contains genuine speech recordings, AI-generated speech samples, and several types of spoofing attacks

Several traditional machine learning algorithms were evaluated and compared, including Random Forest, Support Vector Machine (SVM), and Decision Tree classifiers. Experimental results showed that the Random Forest model achieved the best overall performance among the evaluated models. In addition, audio features such as MFCC proved highly effective for spoofed speech detection tasks

The results demonstrated that the proposed system was able to detect a high percentage of spoofed speech samples, achieving an accuracy exceeding approximately 95% in some experiments, with slight variations depending on the dataset and attack type

One of the main strengths of this study is the simplicity of its methodology and its reliance on computationally efficient traditional machine learning algorithms, making it suitable for low-resource environments. The study also demonstrated the effectiveness of conventional handcrafted speech features for voice deepfake detection. However, one of its major limitations is its dependence on limited handcrafted features, while the performance of traditional machine learning models may significantly decrease when dealing with advanced and unseen deep generative spoofing attacks

The research focused on the main challenge of distinguishing genuine speech from deepfake speech, especially due to the significant improvement in the quality of synthetic speech generation systems. The study emphasized several major challenges, including the strong similarity between real and fake speech characteristics, the diversity of speech generation techniques, and the difficulty of extracting discriminative features capable of identifying spoofed speech

The proposed methodology relied on extracting several audio features from speech recordings, including MFCC, spectral features, and energy-based features. These representations were designed to capture important speech characteristics such as tone, frequency distribution, signal energy, and waveform patterns in order to support machine learning models in distinguishing between genuine and spoofed speech. The study [4] explored Deepfake Voice technology from technical, security, and ethical perspectives. It focused on how synthetic speech is generated using modern artificial intelligence techniques, how fake speech can be detected, and the associated security and ethical risks. The study emphasized that recent advances in deep learning models have made it possible to imitate human voices with extremely high realism, leading to serious threats such as financial fraud, identity impersonation, misinformation dissemination, and political manipulation

The primary objectives of the study were to explain the mechanisms used for generating fake speech, review major voice deepfake detection techniques, and analyze the security, regulatory, and ethical implications of these technologies. The study also highlighted the challenges faced by spoofing detection systems due to the continuous improvement of speech generation models. Several synthetic speech generation architectures were reviewed, including Generative Adversarial Networks (GANs), Autoencoder-based models, Neural Voice Cloning techniques, and advanced Text-To-Speech models such as Tacotron, WaveNet, FastSpeech, and Glow-TTS. The study explained that these systems are capable of accurately

imitating human voice characteristics including tone, pitch, and accent, even using only a few seconds of speech recordings

The paper also discussed both positive and harmful applications of deepfake voice technology. Positive applications include entertainment production, restoring voices for individuals who lost the ability to speak, virtual character generation, and intelligent voice assistant systems. On the other hand, the study highlighted dangerous applications such as financial scams, identity theft, and the generation of fake political or executive voice recordings

Regarding deepfake voice detection methods, the study reviewed several approaches combining signal processing and deep learning. These methods included Spectrogram analysis using Convolutional Neural Networks (CNNs), temporal sequence analysis using recurrent architectures such as RNN and LSTM, and feature-based approaches relying on MFCC features combined with traditional classifiers such as SVM and Decision Trees. The study concluded that hybrid systems integrating signal processing techniques with deep learning models generally achieve the best detection performance

The research methodology consisted of multiple stages. First, speech datasets such as LibriSpeech, VoxCeleb, and ASVspoof were collected, containing both genuine and spoofed speech recordings. Preprocessing steps including noise reduction, silence removal, and volume normalization were then applied. Audio features such as MFCC, Log Mel Spectrogram, and Chroma Features were subsequently extracted

Several deep learning architectures including CNN, LSTM, and Transformer models were trained to classify speech samples as either real or fake. The models were evaluated using multiple performance metrics including Accuracy, Precision, Recall, F1-Score, and ROC-AUC

One of the major strengths of this study is its comprehensive coverage of technical, security, ethical, and regulatory aspects of deepfake voice technology. It also provided a broad overview of both speech generation and spoofing detection techniques. However, the study mainly focused on theoretical and survey-based analysis without proposing a novel practical framework or reporting detailed experimental performance on a specific implemented system. study [5] addressed the problem of voice deepfake detection caused by the rapid advancement of AI-based speech generation technologies such as Voice Cloning, Text-to-Speech (TTS), and Voice Conversion. These technologies have made it increasingly difficult to distinguish between genuine human speech and AI-generated fake speech. The study highlighted the growing risks associated with such technologies, including financial fraud, identity impersonation, and misinformation dissemination

The primary objective of the research was to develop a hybrid deep learning model capable of detecting spoofed speech with high accuracy by exploiting both spectral and temporal characteristics of audio signals. The study also aimed to improve detection performance compared to traditional approaches and evaluate the robustness of the proposed model. The proposed methodology relied on combining three neural network architectures into a unified framework: Convolutional Neural Networks (CNN), Long Short-Term Memory networks (LSTM), and Gated Recurrent Units (GRU). This hybrid architecture was designed to benefit from CNN's ability to extract spectral features while leveraging LSTM and GRU for temporal sequence analysis within speech signals

The proposed system began by processing audio recordings labeled as either real or fake. Several important audio features were extracted, including MFCC, Chroma STFT, RMS Energy, Spectral Centroid, Spectral Bandwidth, Spectral Roll-off, and Zero Crossing Rate. These features helped represent the underlying acoustic patterns and capture subtle differences between genuine and synthesized speech

The preprocessing stage included data standardization, label encoding for converting categorical labels into numerical values, and reshaping operations to prepare the data for CNN input layers. The dataset was then divided into training and testing subsets for proper model evaluation

The model architecture consisted of multiple CNN layers with different convolutional filters combined with ReLU activation, Batch Normalization, MaxPooling, and Dropout techniques to improve learning stability and reduce overfitting. An LSTM layer was subsequently employed to analyze temporal speech sequences and capture patterns related to pitch, rhythm, and temporal variations. A GRU layer was then added to further improve temporal modeling and mitigate the vanishing gradient problem. Finally, a Dense output layer with Sigmoid activation was used for binary classification between real and fake speech

The study evaluated the proposed model using several performance metrics, including Accuracy, Precision, Recall, F1-Score, Equal Error Rate (EER), Confusion Matrix analysis, and K-Fold Cross Validation for more reliable performance assessment

Experimental results demonstrated that the proposed hybrid architecture achieved a high classification accuracy of approximately 99.22%, indicating the effectiveness of combining CNN, LSTM, and GRU for voice deepfake detection. The findings also showed that integrating spectral and temporal feature analysis significantly improved performance compared to traditional machine learning approaches and standalone deep learning models

One of the major strengths of this study is its powerful hybrid architecture that combines spectral and temporal speech analysis, along with the use of multiple advanced evaluation metrics to ensure reliable performance assessment. However, the proposed model has relatively high computational complexity and may require substantial computational resources and longer training times when applied to large-scale datasets or real-time deployment scenarios

Table 1 :Comparative Table of Related Studies :

Study	Year	Techniques Used	Model Used	Dataset	Evaluation Metrics	Main Results
Study [1]	2025	Spectral Features (Mel-Spectrogram, MFCC, CQCC)	CNN, ResNet, CNN+GRU, GMM, PLDA	ASVspoof2019	EER, t-DCF	ResNet + CQCC achieved EER = 0.59%
Study [2]	2025	Mel-Spectrogram, MFCC, CQCC, Spectral Contrast, Chroma Features	CNN, ResNet, CNN+GRU, CNN+LSTM, GMM, SVM	ASVspoof2019, ASVspoof2021	EER, t-DCF	ResNet outperformed traditional CNN models

Study [3]	2024	MFCC, Spectral Features, Energy Features	Random Forest, SVM, Decision Tree	ASVspoof Dataset	Accuracy	Random Forest achieved accuracy above 95%
Study [4]	2025	MFCC, Log Mel Spectrogram, Chroma Features	CNN, LSTM, Transformer	LibriSpeech, VoxCeleb, ASVspoof	Accuracy, Precision, Recall, F1-Score, ROC-AUC	Hybrid deep learning methods improved fake voice detection
Study [5]	2025	MFCC, Chroma STFT, RMS Energy, Spectral Features	CNN + LSTM + GRU	Multiple Audio Datasets	Accuracy, Precision, Recall, F1-Score, EER	Hybrid model achieved 99.22% accuracy

Chapter Four: Research Methodology :

4.1 Introduction :

This research adopts a practical methodology that combines audio signal processing with deep learning techniques to develop a reliable and accurate voice deepfake detection system. The proposed methodology begins with collecting and preprocessing audio data, followed by converting audio signals into suitable spectral representations for analysis. After that, a deep learning model is trained to distinguish between genuine and fake speech based on spectral and temporal features extracted from audio signals. The research methodology also includes performance evaluation and result analysis using several statistical metrics .specialized in classification and voice spoofing detection systems

4.2 Traditional Methods for Solving the Proposed Problem :

Before the emergence of deep learning techniques, voice deepfake detection systems mainly relied on traditional approaches based on handcrafted audio feature extraction and classical machine learning algorithms for classification. These systems analyzed audio signals and extracted statistical or spectral features believed to be capable of distinguishing genuine speech from synthesized speech, after which these features were passed to traditional .classifiers such as Support Vector Machine (SVM) and Gaussian Mixture Model (GMM)

One of the most widely used traditional features is Mel-Frequency Cepstral Coefficients (MFCC), which analyzes speech frequencies in a way that simulates human auditory .perception

Other commonly used features include Linear Predictive Coding (LPC), spectral features, and pitch-related features for analyzing differences between natural and synthesized speech signals. These features were manually designed based on prior knowledge of speech signal .characteristics

At the classification stage, traditional algorithms such as SVM, Random Forest, and Hidden Markov Models (HMM) were widely used to build systems capable of distinguishing between genuine and fake speech. Although these methods were computationally efficient and relatively simple compared to modern approaches, they suffered from several limitations. Their performance heavily depended on the quality of manually extracted features, and they lacked the ability to capture complex patterns and subtle artifacts .generated by advanced speech synthesis technologies

With the rapid advancement of AI-based voice generation techniques, traditional methods became less effective in detecting highly realistic fake speech due to their limited generalization capability and inability to learn hidden representations directly from raw audio data. Consequently, recent research has shifted toward deep learning approaches, which can automatically learn discriminative features from data without requiring manual feature engineering, leading to significantly improved performance in voice deepfake detection tasks

4.3 Numerical Methods :

This project relies on several numerical methods and mathematical algorithms used in data processing and deep learning model training. These methods play a fundamental role in improving model performance and minimizing error during the learning process. Numerical techniques are applied in multiple stages of the system, starting from audio signal processing and feature extraction to neural network training and parameter optimization.

One of the most important numerical operations used in the project is the Fast Fourier Transform (FFT), which is a fundamental mathematical algorithm for converting audio signals from the time domain into the frequency domain. This transformation decomposes speech signals into different frequency components in order to extract important spectral characteristics that help distinguish genuine speech from fake speech. In addition, the Short-Time Fourier Transform (STFT) is applied by dividing the signal into small overlapping windows and analyzing each segment independently, allowing both temporal and spectral information to be preserved simultaneously.

The project also relies on numerical normalization techniques that standardize input values and reduce variations between audio samples, which improves training stability and accelerates model convergence. Furthermore, padding and trimming operations are used to unify the dimensions of Mel-Spectrogram representations so that all samples have fixed input sizes before being fed into the neural network.

During the training phase, the project employs the Backpropagation algorithm, which calculates the partial derivatives of the loss function with respect to the neural network parameters. This process is based on gradient computation to determine how the network weights should be updated in order to minimize classification error and improve performance iteratively.

4.4 Machine Learning Methods :

This project relies on Machine Learning and Deep Learning techniques, which are among the most important branches of artificial intelligence used to develop systems capable of learning from data and discovering hidden patterns within it. Machine learning methods aim to build models that can automatically analyze data, make decisions, and perform classification tasks based on knowledge acquired during the training process.

In the early stages of voice deepfake detection systems, traditional machine learning algorithms such as Support Vector Machine (SVM), Random Forest, and Gaussian Mixture Models (GMM) were widely used. These models relied on handcrafted audio features such as MFCC and spectral features extracted manually from speech signals. Although these approaches achieved acceptable performance on simpler datasets, they struggled to handle the complex patterns generated by modern deepfake synthesis techniques, and their effectiveness strongly depended on the quality of manually engineered features.

4.5 Proposed Model Architecture, Loss Function, and Training Algorithms :

The proposed model in this research is based on a hybrid architecture that combines Convolutional Neural Networks (CNNs) and Long Short-Term Memory (LSTM) networks in order to benefit from the strengths of both approaches in analyzing speech signals and detecting voice deepfakes. The model is designed to process Mel-Spectrogram representations extracted from audio recordings after applying preprocessing stages such as .resampling, mono conversion, silence removal, and audio normalization

The architecture begins by receiving fixed-size Mel-Spectrogram inputs representing the temporal and spectral characteristics of speech signals

These inputs are then passed through several CNN layers responsible for extracting spectral features and local patterns from the spectrogram images. The convolutional layers utilize multiple filters to detect artifacts and subtle differences between genuine and synthesized speech signals. In addition, ReLU activation functions are applied to introduce non-linearity into the network, while Max Pooling layers are used to reduce dimensionality, retain the .most informative features, and decrease computational complexity

After the feature extraction stage, the generated feature maps are passed to LSTM layers, which are responsible for analyzing temporal dependencies within the speech signal. LSTM networks are capable of preserving important sequential information over time and effectively addressing the Vanishing Gradient Problem commonly associated with traditional recurrent neural networks. This enables the model to better understand speech dynamics .and temporal variations found in both real and fake audio recordings

In the final stage, the extracted features are fed into Fully Connected Layers to perform binary classification between genuine and fake speech samples. A Sigmoid activation function is used in the output layer to convert the model outputs into probability values ranging from 0 to 1, where values closer to 1 indicate fake speech and values closer to 0 .indicate genuine speech

Regarding the loss function, Binary Cross Entropy Loss (BCE Loss) was selected because the problem is formulated as a binary classification task. This loss function measures the difference between the predicted outputs and the actual labels, where lower loss values indicate better classification performance. In addition, a weighted version of the loss function was employed to address the data imbalance problem between genuine and fake samples by assigning higher importance to the minority class, improving the model's ability .to detect spoofed speech

The training process relies on the Backpropagation algorithm, which computes gradients for the neural network parameters based on the calculated loss values. Model parameters are then updated using the Adam Optimizer, which is one of the most widely used optimization algorithms in deep learning due to its efficiency, fast convergence, and training stability. Adam utilizes adaptive moving averages of previous gradients to improve weight updates .and accelerate optimization

Furthermore, Mini-Batch Training is applied during training by dividing the dataset into smaller batches instead of processing the entire dataset at once. This improves computational efficiency and reduces memory consumption. Finally, the model performance

is evaluated after each training epoch using several metrics such as Accuracy, Recall, F1-Score, and Equal Error Rate (EER) to monitor learning progress and assess the generalization capability of the proposed system

4.6 Detailed Pipeline Explanation :

The proposed voice deepfake detection system is based on a sequential processing pipeline that starts from raw audio file loading and ends with generating a classification decision indicating whether the speech recording is genuine or fake. The pipeline was designed to ensure proper audio preprocessing, stable model training, and effective extraction of discriminative speech features before feeding data into the deep learning architecture

The first stage begins with reading dataset protocol files and metadata, where audio file paths and their corresponding labels are extracted. Each sample is labeled as either genuine or fake. Afterward, textual labels are converted into numerical values, where 0 represents genuine speech and 1 represents fake speech, enabling their use during neural network training

In the second stage, audio files are loaded using the Torchaudio library, which extracts both the waveform and the sampling rate of each audio recording. Since the dataset contains recordings with different characteristics, several preprocessing steps are applied to standardize and improve the quality of the input signals

The preprocessing pipeline starts with converting audio recordings into mono channels whenever multiple channels are detected. This is achieved by averaging the audio channels into a single waveform, reducing computational complexity and ensuring consistent input structure. After that, a resampling process is performed to standardize all recordings to a sampling rate of 16 kHz, ensuring that all speech samples contain the same temporal resolution and number of samples per second

Once the sampling rate is unified, silence removal is applied using the Librosa library to eliminate silent or low-energy regions from audio recordings. This step reduces unnecessary information and allows the model to focus on meaningful speech content. Next, audio normalization is performed by dividing waveform values by the maximum absolute amplitude value, ensuring consistent signal ranges and improving neural network training stability

The processed audio signals are then converted into Mel-Spectrogram representations using the Torchaudio library. This transformation relies on the Short-Time Fourier Transform (STFT) to analyze frequency variations over time, followed by projecting frequencies into the Mel scale, which simulates human auditory perception. A total of 128 Mel frequency bands were used to obtain rich spectral representations capable of capturing patterns associated with genuine and synthesized speech

After generating the Mel-Spectrogram, logarithmic transformation is applied to convert signal energy values into the logarithmic domain, improving feature visibility and reducing large energy variations within the spectrogram. The unnecessary channel dimension is then removed using the squeeze operation, producing a two-dimensional representation suitable for neural network processing

Since audio recordings have different durations, a fixed-length processing stage is applied using padding and trimming operations. If the Mel-Spectrogram length is shorter than the predefined maximum length, zero-padding is added at the end of the spectrogram.

Otherwise, excessive frames are trimmed. This step ensures that all input samples have identical dimensions before being passed to the neural network

After preprocessing, a custom Dataset object is implemented using PyTorch to associate each audio sample with its corresponding label. DataLoader objects are then used to divide the dataset into mini-batches containing 16 samples each. Shuffle mode is enabled during training to randomize sample ordering and reduce learning bias

During the training stage, Mel-Spectrogram inputs are passed into the CNN-LSTM architecture

CNN layers are responsible for extracting spectral patterns and local features from spectrogram representations, while LSTM layers analyze temporal dependencies and sequential relationships between speech frames. The extracted features are then passed through Fully Connected layers to perform final binary classification using the Sigmoid activation function

During optimization, Binary Cross Entropy Loss with class weighting is used to address dataset imbalance problems. Backpropagation is then applied to compute gradients and update model parameters using the Adam optimizer. Model performance is evaluated after each epoch using Accuracy, Recall, F1-Score, and Equal Error Rate (EER) metrics to monitor learning progress and assess the system's capability in detecting voice deepfakes

4.7 Datasets Used :

This project relies on a voice deepfake detection dataset containing both genuine and synthesized speech recordings generated using artificial speech synthesis techniques. The dataset was utilized to train a deep learning model capable of distinguishing between real and fake speech based on spectral and temporal characteristics extracted from audio signals. The dataset includes multiple audio files representing different speech samples, where each file is associated with a label indicating whether the recording is genuine or fake

The dataset was divided into two main subsets: a Training Set and a Validation/Development Set. The training data was used to teach the model the discriminative patterns within audio signals, while the validation set was employed to evaluate model performance during training and measure its generalization capability on unseen samples

The dataset exhibited a class imbalance problem, where the number of genuine samples was significantly larger than the number of fake samples. To address this issue, a weighted loss function was applied during training, assigning higher importance to the minority class in order to improve the model's ability to detect spoofed speech and reduce classification bias toward the majority class

Before feeding the data into the model, several preprocessing stages were applied to improve audio quality and standardize signal characteristics. The preprocessing pipeline started with converting audio recordings into mono channels whenever multi-channel audio

was detected, reducing computational complexity and ensuring input consistency. After that, a resampling process was performed to unify the sampling rate at 16 kHz, which is widely adopted in speech processing and deep learning applications

In addition, silence removal was applied using the Librosa library to eliminate unnecessary silent segments from audio recordings, enabling the model to focus on meaningful speech information. Audio normalization was then performed to standardize signal amplitudes and reduce variations between samples. After preprocessing, audio signals were transformed into Mel-Spectrogram representations using the Torchaudio library. This representation converts speech into a two-dimensional image describing the relationship between time, frequency, and signal energy

To ensure compatibility with the neural network architecture, Mel-Spectrogram dimensions were standardized using padding and trimming operations so that all samples had fixed input sizes before training. Several tools and libraries were used throughout the project, including PyTorch, Torchaudio, Librosa, and NumPy for preprocessing, feature extraction, and model development

Examples of the utilized data include genuine human speech recordings alongside synthesized speech samples generated using AI-based voice cloning and speech synthesis technologies, allowing the model to learn subtle differences between natural and artificially generated speech patterns

4.8 Chapter Conclusion and Research Contribution :

This chapter presented the methodology adopted for developing the proposed voice deepfake detection system, starting from the theoretical foundations of audio signal processing and machine learning techniques to the detailed explanation of the proposed model architecture, preprocessing stages, and training procedures. The chapter also explained the role of Mel-Spectrogram representations and CNN-LSTM networks in extracting spectral and temporal features and analyzing speech patterns associated with genuine and synthesized audio signals

In addition, traditional approaches previously used for voice spoofing detection were discussed along with their limitations compared to modern deep learning models. Numerical methods and optimization algorithms used during training were also presented, followed by a detailed explanation of the utilized datasets and the preprocessing pipeline applied before feeding data into the neural network

The main research contribution of this project lies in proposing a hybrid architecture that combines spectral and temporal feature analysis to improve the effectiveness of voice deepfake detection systems. The project also contributes by developing a complete audio preprocessing pipeline and addressing the data imbalance problem using a weighted loss function. Furthermore, advanced evaluation metrics such as Equal Error Rate (EER) were adopted to provide a more realistic assessment of model performance. The proposed methodology establishes a scalable foundation for future improvements through the integration of modern approaches such as Self-Supervised Learning to enhance robustness and generalization capability against unseen voice spoofing attacks

Chapter Five: Research Results :

5.1 Tools Used and Evaluation Methods :

This project was implemented using several programming tools and specialized libraries related to deep learning and audio signal processing. The Python programming language was selected due to its widespread use in artificial intelligence and data analysis applications. In addition, Google Colab was utilized to provide GPU acceleration and improve the efficiency of model training and experimentation.

The project mainly relied on the PyTorch library for building the deep learning model and performing training and evaluation processes because of its flexibility and strong support for modern neural network architectures. The Torchaudio library was used for audio processing tasks such as resampling and converting audio signals into Mel-Spectrogram representations. Furthermore, the Librosa library was employed for silence removal and audio signal analysis, while NumPy was used for numerical operations and data manipulation.

For visualization and result analysis, Matplotlib was utilized to plot training and loss curves in order to monitor model performance during the learning process. In addition, the Pandas library was used for organizing dataset files and handling protocol files associated with speech labels and metadata.

The proposed model was evaluated using several statistical metrics specialized in binary classification and voice deepfake detection tasks. One of the primary evaluation metrics was Accuracy, which measures the overall percentage of correctly classified samples. Recall was also used to evaluate the model's ability to correctly identify fake speech samples, which is particularly important in spoofing and fraud detection systems.

Moreover, the F1-Score metric was adopted to provide a balanced evaluation by combining Precision and Recall into a single measurement.

In addition, Equal Error Rate (EER) was used as one of the most important evaluation metrics in biometric authentication and voice spoofing detection systems. EER measures the point at which the False Acceptance Rate equals the False Rejection Rate, and lower EER values indicate better system reliability and performance.

During training, Binary Cross Entropy Loss was used to monitor the learning process and measure the difference between predicted outputs and actual labels. Loss values were tracked across multiple epochs to analyze training stability and convergence behavior. Training and validation curves were also plotted to monitor overfitting and evaluate the model's generalization capability on unseen data.

5.2 Proposed Model Results :

After applying the preprocessing stages and training the proposed CNN-LSTM hybrid architecture, the experimental results demonstrated the model's strong ability to distinguish between genuine and fake speech recordings. The model was trained over multiple epochs while monitoring both training and validation losses in order to analyze training stability and .evaluate generalization capability

The training process showed a gradual reduction in loss values, where the Training Loss decreased from 0.6675 at the beginning of training to 0.0552 at the final epoch. Similarly, the Validation Loss decreased from 0.5372 to 0.2084, indicating that the model successfully learned discriminative speech patterns and improved its ability to classify genuine and fake .audio samples

The proposed model also achieved high classification accuracy, reaching approximately 99.67% on the training set and 98.09% on the validation set. These results indicate strong .learning capability and good generalization performance on unseen data

Regarding additional evaluation metrics, the model achieved a Recall value of 97.21% in the best experiment. Recall is considered a critical metric in voice deepfake detection systems because it measures the ability of the model to correctly identify fake speech samples while minimizing missed spoofing attacks. Furthermore, the model achieved an F1-Score of 62.62%, reflecting the balance between Precision and Recall under the class imbalance .conditions present in the dataset

In addition, the Equal Error Rate (EER) metric was used to evaluate system reliability. The proposed model achieved an EER value of 0.0690, which is considered relatively low and indicates reduced false acceptance and false rejection rates, leading to improved reliability .in detecting spoofed speech

These results demonstrate that combining Mel-Spectrogram representations with the CNN-LSTM hybrid architecture effectively captured both spectral and temporal characteristics .associated with voice deepfakes

5.3 Results Comparison :

Several experiments were conducted to evaluate the performance of the proposed model and analyze its effectiveness in detecting voice deepfakes. The comparison process focused on studying the impact of preprocessing techniques, model architecture, and evaluation metrics used during experimentation. In addition, the proposed system was compared with traditional approaches and previously published methods in order to identify its strengths .and limitations

The results demonstrated that using Mel-Spectrogram representations combined with deep neural networks achieved superior performance compared to traditional methods based on handcrafted features such as MFCC combined with classifiers like SVM or GMM. This improvement is mainly attributed to the ability of deep learning models to automatically

learn complex patterns and subtle artifacts hidden within synthesized speech signals without .relying on manually engineered features

The experiments also showed that combining CNN layers with LSTM layers improved performance compared to using CNN architectures alone. CNN layers were effective in extracting local spectral features from spectrogram representations, while LSTM layers successfully captured temporal dependencies and sequential variations within speech signals. This hybrid architecture enhanced the model's capability to detect sophisticated .voice spoofing attacks and high-quality synthesized speech samples

Furthermore, the use of a Weighted Loss Function significantly improved the model's ability to handle the dataset imbalance problem. In particular, Recall values for the fake speech class improved compared to experiments that did not utilize class weighting. This is especially important in spoofing detection systems because failing to detect fake samples may lead to serious security risks

Regarding evaluation metrics, the experiments revealed that relying solely on Accuracy is insufficient due to dataset imbalance. Therefore, additional metrics such as Recall, F1-Score, and Equal Error Rate (EER) were used to provide a more reliable assessment of model performance. The relatively low EER value indicated that the proposed system effectively .reduced both false acceptance and false rejection rates

Overall, the comparison results confirmed that the proposed model is effective for voice deepfake detection tasks and that the combination of suitable preprocessing techniques with the CNN-LSTM hybrid architecture contributed significantly to improving detection .performance compared to traditional methods and simpler baseline models

5.4 Comparison of Proposed Models Results :

In this research, two different approaches for voice deepfake detection were implemented and evaluated. The first model was based on a hybrid CNN-LSTM architecture, while the second model relied on Self-Supervised Learning using advanced speech representations .extracted from a Wav2Vec2-based framework

The first model utilized Mel-Spectrogram representations as the primary input to the neural network. CNN layers were responsible for extracting spectral and local speech features, while LSTM layers analyzed temporal dependencies within speech signals. The training results showed a gradual decrease in loss values, where the Training Loss decreased from 0.6675 at the beginning of training to 0.0552 at the final epoch. Similarly, the Validation Loss decreased from 0.5372 to 0.2084, indicating successful feature learning and improved .classification capability

The CNN-LSTM model also achieved very high classification performance, reaching approximately 99.67% Training Accuracy and 98.09% Validation Accuracy. In addition, the model achieved a Recall value of approximately 84.85%, demonstrating strong capability in .detecting spoofed speech samples while maintaining stable learning behavior

The second model, based on Self-Supervised Learning using Wav2Vec2 representations, demonstrated superior capability in extracting deep and meaningful speech representations

directly from raw audio signals without relying on handcrafted features. Experimental results showed consistent improvement during training, where the Training Loss decreased from 0.4723 to 1848, while the Validation Loss decreased from 0.4320 to 0.3065 over three .training epochs

The self-supervised model achieved particularly strong Recall performance, reaching 98.70% in the best experiment, indicating a very high capability in detecting spoofed speech attacks and minimizing missed fake samples. Moreover, the F1-Score improved progressively and reached 62.62% in the final epoch, while the Equal Error Rate (EER) decreased to 0.0690, .demonstrating improved system reliability and lower false acceptance and rejection rates

The final comparison revealed that the CNN-LSTM model achieved higher training stability and overall classification accuracy while requiring lower computational resources and shorter training time compared to the self-supervised model. On the other hand, the Self-Supervised Learning model achieved better Recall and EER performance due to its ability to .leverage advanced pretrained speech representations learned from large-scale datasets

Based on the obtained results, the CNN-LSTM architecture can be considered more suitable for applications requiring computational efficiency and faster deployment, whereas the Self-Supervised Learning approach is more appropriate for security-critical applications that demand stronger spoofing detection capability and lower risk of undetected fake speech attacks

5.5 Comparison with Previous Works :

When comparing the proposed models with previous studies in the field of voice deepfake detection, the obtained results can be considered competitive with many modern systems, .particularly in terms of Recall and Equal Error Rate (EER) performance

Many previous studies relied on traditional CNN-based architectures or handcrafted audio features such as MFCC, where Recall values typically ranged between 85% and 95%, while EER values commonly ranged from 0.10 to 0.20 depending on the dataset and spoofing attack types. Although these systems achieved acceptable performance, they often struggled to detect high-quality synthesized speech generated by modern AI-based voice .synthesis techniques

In this project, the first CNN-LSTM model achieved a high Validation Accuracy of 98.09% along with a Recall value of approximately 84.85%, demonstrating strong classification .capability and stable training behavior with progressively decreasing loss values

The second model based on Self-Supervised Learning using Wav2Vec2 achieved results closer to recent Transformer-based deepfake detection systems. The model reached a Recall value of 98.70% while reducing the Equal Error Rate to 0.0690, which is considered superior to many traditional approaches reported in previous studies. These results indicate that pretrained speech representations significantly improved the model's ability to learn deep .and discriminative speech features

The comparison also showed that preprocessing techniques such as silence removal, normalization, sampling rate standardization, and the use of weighted loss functions

contributed significantly to improving model performance compared to some previous studies that relied on simpler preprocessing pipelines

Although some recent large-scale Transformer-based systems may achieve slightly better performance on certain benchmarks, they typically require substantially higher computational resources and longer training times. In contrast, the proposed system achieved a strong balance between performance, computational efficiency, and practical deployment capability

Overall, the obtained results demonstrate that the proposed system is capable of competing with modern voice deepfake detection approaches, especially when Self-Supervised Learning techniques are employed. These findings also provide a solid foundation for future improvements through deeper fine-tuning strategies and the use of larger and more diverse speech datasets

Chapter Six: Conclusion :

6.1 Conclusions :

This research addressed the problem of voice deepfake detection as one of the most critical modern challenges in cybersecurity and artificial intelligence. Recent advances in AI-based speech synthesis technologies have made it possible to generate highly realistic fake speech recordings that are difficult to distinguish from genuine human voices. To address this issue, a deep learning-based system was developed for detecting spoofed speech with high accuracy and reliability.

The proposed framework relied on several stages beginning with audio preprocessing, including mono conversion, resampling, silence removal, normalization, and Mel-Spectrogram generation for extracting important spectral and temporal speech features. Two different models were then developed and evaluated: a hybrid CNN-LSTM architecture and a Self-Supervised Learning approach based on Wav2Vec2 representations.

Experimental results demonstrated that the CNN-LSTM model achieved high classification accuracy and stable training behavior, while the Self-Supervised Learning model showed superior capability in detecting spoofed speech and achieved better Recall and Equal Error Rate (EER) performance. The results also confirmed that pretrained speech representations and effective preprocessing techniques play a significant role in improving the performance of voice deepfake detection systems.

Furthermore, the study showed that relying on multiple evaluation metrics such as Recall, F1-Score, and EER provides a more realistic assessment of model performance compared to using Accuracy alone, especially in imbalanced datasets. Overall, the findings indicate that combining modern deep learning approaches with efficient speech preprocessing pipelines can produce robust and scalable systems suitable for future biometric security and anti-spoofing applications.

6.2 Challenges and Future Work :

Several technical and research challenges were encountered during the development of the proposed voice deepfake detection system. One of the major challenges was the class imbalance problem, where the number of genuine speech samples was significantly larger than the number of fake samples. Although this issue was partially addressed using a weighted loss function, dataset imbalance remains a major challenge in modern spoofing detection systems.

Another challenge was the high computational cost associated with deep learning models, particularly Self-Supervised Learning architectures such as Wav2Vec2, which require substantial computational resources and longer training times compared to traditional models. Furthermore, the rapid evolution of speech synthesis technologies makes it increasingly difficult to develop systems capable of detecting all future spoofing attacks with complete reliability.

Additional challenges include variations in recording quality, environmental noise, speech styles, and accents, all of which can affect model performance and reduce generalization capability on real-world unseen data

Regarding future work, the proposed system can be improved by utilizing larger and more diverse datasets containing multiple languages and accents. In addition, deeper fine-tuning of Self-Supervised Learning models could further enhance the quality of extracted speech representations. Modern Transformer architectures and multimodal approaches combining audio and visual information may also be integrated to improve system robustness and detection reliability

Future improvements may also focus on optimizing the system for real-time deployment, enabling its use in cybersecurity applications, biometric authentication systems, call centers, and social media platforms for early detection of voice spoofing and audio-based fraud attacks

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